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Measurement device for characterizing a device-under-test

Abstract

The present disclosure relates to a measurement device for characterizing a device-under-test, DUT. The device includes an RF signal generator to generate test signals in parallel. The test signals each have a different frequency. A signal path connects the RF signal generator to a port of the measurement device. The port is connected to the DUT. The signal path feeds test signals to the port and receives response signals of the DUT from the port. The device includes a measurement unit; a forward coupler connected to the signal path and forwards at least a part of each of the test signals to the measurement unit; and a reverse coupler is connected to the signal path and forwards at least a part of the response signals to the measurement unit. The measurement unit simultaneously measures the forwarded parts of the test and response signals, each in amplitude and phase.

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Background/Summary

TECHNICAL FIELD

(1) In general, the disclosure relates to the characterization and testing of RF systems and components. More specifically, the present disclosure relates to a measurement device, such as a vector network analyzer, for characterizing a device-under-test (DUT).

BACKGROUND ART

(2) RF (radio frequency) equipment, such as signal amplifiers or passive components (e.g., cables), can be characterized and/or tested by supplying them with signals of different frequencies and analyzing how these signals are modified or distorted by the equipment. This often involves a large number of successive measurements with different signals, which can lead to long measurement times.

(3) For example, when characterizing Ethernet cables according to the IEEE803 standard, so-called 4-port measurements are repeatedly performed on the passive lines of the cables. In order to characterize all influences caused by crosstalk, several hundred 4-port measurements are necessary, which add up to several dozen minutes or even hours.

SUMMARY

(4) Thus, there is a need to provide an improved measurement device for characterizing a DUT, which avoids the above-mentioned disadvantages. In particular, the measurement times to characterize the DUT should be reduced.

(5) This is achieved by the embodiments provided in the enclosed independent claims. Advantageous implementations of the present disclosure are further defined in the dependent

claims.
(6) According to a first aspect, the present disclosure relates to a measurement device for characterizing a device-under-test, DUT. The measurement device comprises: an RF signal generator configured to generate at least two test signals in parallel, wherein the at least two test signals each have a different frequency; a signal path connecting the RF signal generator to a port of the measurement device, wherein the port is arranged for being connected to the DUT; the signal path being configured to feed the at least two test signals to the port and to receive at least two response signals of the DUT from the port; a measurement unit; a forward coupler being connected to the signal path and adapted to forward at least a part of each of the at least two test signals to the measurement unit; and a reverse coupler being connected to the signal path and adapted to forward at least a part of each of the at least two response signals to the measurement unit. The measurement unit is adapted to simultaneously measure the forwarded parts of the at least two test signals and the at least two response signals, each in amplitude and phase.
(7) This achieves the advantage that measurement times needed to characterize the DUT are reduced. This is, in particular, due to the parallelization of the measurements with multiple signals at different frequencies. For instance, when measuring with two test respectively response signals in parallel, the measurement time can be reduced by a factor of two compared to measuring with only one test signal at a time.
(8) The at least two test signals can be RF signals. In particular, the test signals are tones, i.e. signals with a defined frequency and a narrow bandwidth. Likewise, the response signals can also be RF signals and/or tones.
(9) The test signals can be CW signals with different frequencies. For instance, the at least two test signals can comprise at least a first test signal at a first frequency and at least a second test signal at a second frequency different to the first frequency.
(10) The response signals can be generated by the DUT in response to receiving the test signals. For instance, the at least two response signals can comprise at least a first response signal and at least a second response signal. The first response signal can be generated by the DUT in response to the first test signal and the second response signal can be generated by the DUT in response to the second test signal. The response signals can be CW signals and/or can have different frequencies.
(11) In particular, the test signals are travelling from the RF signal generator to the port, while the response signals are travelling from the port to the RF signal generator. As such, the response signals can be reflections of the test signals (i.e., reflected signals based on the test signals) form the DUT.
(12) The measurement device can be a vector network analyzer (VNA).
(13) The DUT can be an active or a passive RF device. For instance, the DUT can be an amplifier or a cable, e.g. an Ethernet cable. The DUT can be connected to the port via a suitable connections means, e.g., an adapter and/or a connecting cable.
(14) The port can be an RF port and can comprise a coaxial RF connector.
(15) The RF signal generator can comprise one or more RF signal sources. For example, the RF signal generator comprises one RF signal source for each test signal which is generated in parallel. Alternatively, the RF signal generator can also comprise a single signal generator capable of generating multiple RF test signals, e.g. a comb generator.
(16) The forward coupler and the reverse coupler can each be directional couplers.
(17) The measurement unit can comprise a processing unit, such as a microprocessor or an ASIC. The measurement unit can comprise at least one integrated circuit.
(18) Here, simultaneously measuring forwarded parts of the test and the response signals may refer to measuring the forwarded parts of these signals in parallel.
(19) In an embodiment, the RF signal generator is configured to amend the frequencies of the at least two test signals by predefined and/or adjustable frequency steps over a determined frequency

bandwidth.
(20) This achieves the advantage that the measurement device can sweep through a specific frequency range. The RF signal generator can amend the frequencies of the test signals stepwise or continuously. In this way, multiple parallel measurements can be carried out consecutively.
(21) In an embodiment, the at least two test signals are each increased by the same frequency, starting from their respective initial frequencies.
(22) In an embodiment, the measurement device further comprises: a first mixing unit which is configured to down-convert the forwarded parts of the at least two test signals into intermediate frequency, IF, signals; and a first analog-to-digital converter, ADC, unit which is configured to digitize the down-converted parts of the test signals; and the measurement unit further comprises: a second mixing unit which is configured to down-convert the forwarded parts of the at least two response signals into IF signals; and a second ADC unit which is configured to digitize the down-converted parts of the response signals.
(23) This achieves the advantage that the test signals and the response signals can be fed to the measurement unit in a digitalized form.
(24) Here, the “down-converted parts” of the test signal(s) respectively response signal(s) refer to the parts of said signals which were forwarded by the respectively couplers and then down-converted by the mixing units. These “down-converted parts” can then be digitalized by the ADC units.
(25) In an embodiment, the first mixing unit comprises only one mixer which is configured to down-convert the forwarded parts of the at least two test signals; and the first ADC unit comprises only one ADC which is configured to digitize the down-converted parts of the test signals.
(26) In an embodiment, the second mixing unit comprises only one mixer which is configured to down-convert the forwarded parts of the at least two response signals; and the second ADC unit comprises only one ADC which is configured to digitize the down-converted parts of the response signals.
(27) For example, the one ADC of the first ADC unit can be a broadband ADC which is suitable to digitalize the at least two test signals in parallel. Likewise, the one ADC of the second ADC unit can also be a broadband ADC which is suitable to digitalize the at least two response signals in parallel.
(28) In an embodiment, the first mixing unit comprises at least two mixers, wherein a first mixer of the at least two mixers is configured to down-convert the forwarded part of a first one of the at least two test signals and wherein a second mixer of the at least two mixers is configured to down-convert the forwarded part of a second one of the at least two test signals; and wherein the first ADC unit comprises at least two ADCs, wherein a first ADC of the at least two ADCs is configured to digitize the down-converted part of the first one of the test signals and wherein a second ADC of the at least two ADCs is configured to digitize the down-converted part of the second one of the test signals.
(29) In an embodiment, the second mixing unit comprises at least two mixers, wherein a first mixer of the at least two mixers is configured to down-convert the forwarded part of a first one of the at least two response signals and wherein a second mixer of the at least two mixers is configured to down-convert the forwarded part of a second one of the at least two response signals; and wherein the second ADC unit comprises at least two ADCs, wherein a first ADC of the at least two ADCs is configured to digitize the down-converted part of the first one of the response signals and wherein a second ADC of the at least two ADCs is configured to digitize the down-converted part of the second one of the response signals.
(30) For example, if the ADC units comprise multiple ADCs, each ADC can be a narrowband ADC suitable for digitalizing one of the test respectively response signals.
(31) In an embodiment, the measurement device further comprises a further signal path connecting the RF signal generator to a further port of the measurement device, wherein the further port is

arranged for being connected to the DUT; and wherein the further signal path is configured to feed the at least two test signals to the further port and/or to receive at least two further response signals of the DUT from the further port.

(32) The at least two further response signals can be signals generated by the DUT in response to the at least to two test signals forwarded to the DUT via the port or via the further port. The at least two further response signals can comprise at least a third and a fourth response signal.

(33) The at least two test signals forwarded by the further signal path can be identical to the at least two test signals forwarded by the signal path (in particular, identical in frequency and amplitude). For example, the measurement device can first forward the at least two test signals to the DUT via the signal path and the port and, subsequently, via the further signal path and the further port.

(34) In an embodiment, the measurement device comprises: a further forward coupler being connected to the further signal path and adapted to forward at least a part of each of the at least two test signals to the measurement unit; and/or a further reverse coupler being connected to the further signal path and adapted to forward at least a part of each the at least two further response signals to the measurement unit.

(35) For example, the measurement unit is adapted to simultaneously measure the forwarded parts of the at least two test signals and/or the at least two further response signals, each in amplitude and phase.

(36) In an embodiment, in a first operating state, the measurement device is configured to transmit the at least two test signals through the first signal path to the first port; and to: a) forward parts of each of the at least two test signals via the forward coupler to the measurement unit; b) forward parts of each of the at least two response signals from the first port via the reverse coupler to the measurement unit; and c) forward parts of each of the at least two further response signals from the further port via the further reverse coupler to the measurement unit.

(37) In an embodiment, in a second operating state, the measurement device is configured to transmit the at least two test signals through the further signal path to the further port; and to: a) forward parts of each of the at least two test signals via the further forward coupler to the measurement unit; b) forward parts of each of the at least two response signals from the port via the reverse coupler to the measurement unit; and c) forward parts of each of the at least two further response signals from the further port via the further reverse coupler to the measurement unit.

(38) In both the first and second operating state, the measurement unit can be adapted to simultaneously measure the forwarded parts of the at least two test signals, the at least two response signals and the at least two further response signals, each in amplitude and phase.

(39) In an embodiment, the measurement unit is configured to calculate S-parameters of the DUT over a determined frequency bandwidth at least based on the measurements of the forwarded parts of the at least two test signals and the at least two response signals.

(40) In an embodiment, the measurement device comprises at least four ports, wherein each of the at least four ports is arranged for being connected to the DUT, wherein each of the at least four ports is connected to the RF signal generator by a respective signal path, wherein the measurement device is adapted to apply the at least two test signals to each of the at least four ports in an alternating manner; and wherein each signal path is connected to the measurement unit by a respective forward coupler and a respective reverse coupler. This achieves the advantage that the measurement device can carry out a 4-port measurement with the DUT.

(41) According to a second aspect, the present disclosure relates to a system comprising: the measurement device of the first aspect of the disclosure; and a switch matrix which is connected between at least one port of the measurement device and the DUT; wherein the measurement device is configured to control the switch matrix.

(42) In an embodiment, the switch matrix comprise a plurality of test ports adapted to be connected to one or more than one DUT; wherein the switch matrix is configured to subsequently connect each of the plurality of the test ports to the at least one port of the measurement device. This

achieves the advantage that DUT, such as cables, can be measured in a simple and efficient way.
(43) According to a third aspect, the present disclosure relates to a method of operating the system of the second aspect of the disclosure to analyze a DUT, like a cable, which comprises a plurality of conductor lines. The method comprises the steps of: connecting both ends of the plurality of conductor lines to respective test ports of the switch matrix; forwarding the at least two test signals to one end of one of the plurality of conductor lines; and measuring the at least two response signals at the other end of the conductor line which receives the at least two test signals, and at both ends of one, more or all further conductor lines of the plurality of conductor lines. Of course, the frequency of the at least two test signals can be swept through a pre-defined frequency band. The response signals can be measured at more or all further conductor lines in parallel or sequentially. If the measurement is done sequentially, then the test signals are preferably applied and swept through a pre-defined frequency band for each of the further conductor lines

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Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The above-described aspects and implementation forms of the present disclosure will be explained in the following description of specific embodiments in relation to the enclosed drawings, in which:
(2) FIG. **1** shows a schematic diagram of a measurement device for characterizing a DUT according to an embodiment;
(3) FIG. **2** shows a schematic diagram of a measurement device for characterizing a DUT according to an embodiment;
(4) FIG. **3** shows a schematic diagram of a measurement device for characterizing a DUT according to an embodiment;
(5) FIGS. **4**A-**4**B show schematic diagrams of a measurement device for characterizing a DUT according to an embodiment;
(6) FIG. **5** shows a schematic diagram of a system comprising a measurement device according to an embodiment; and
(7) FIG. **6** shows a flow diagram of a method of operating a system comprising a measurement device and a switch matrix according to an embodiment.
DETAILED DESCRIPTIONS OF EMBODIMENTS
(8) FIG. **1** shows a schematic diagram of a measurement device **10** for characterizing a DUT **11** according to an embodiment.
(9) The measurement device **10** comprises: an RF signal generator **12** configured to generate at least two test signals in parallel, wherein the at least two test signals each have a different frequency; a signal path **13** connecting the RF signal generator **12** to a port **14** of the measurement device, wherein the port **14** is arranged for being connected to the DUT **11**. The signal path **13** is configured to feed the at least two test signals to the port **14** and to receive at least two response signals of the DUT **11** from the port **14**. The measurement device **10** further comprises a measurement unit **15**; a forward coupler **16** being connected to the signal path **13** and adapted to forward at least a part of each of the at least two test signals to the measurement unit **15**; and a reverse coupler **17** being connected to the signal path **13** and adapted to forward at least a part of each of the at least two response signals to the measurement unit **15**. The measurement unit **15** is adapted to simultaneously measure the forwarded parts of: the at least two test signals and the at least two response signals, each in amplitude and phase.
(10) The DUT **11** can be an active RF device, such as an amplifier, or a passive device, such as a cable.
(11) Each of the test signals and/or the response signals can be a CW signal (i.e., a sine signal). The

at least two test signals can comprise a first test signal and a second test signal, and the at least two response signals can comprise a first response signal and a second response signal. For example, the first response signal can be generated by the DUT based on the first test signal, wherein it has the same frequency as the first test signal but differs from the first test signal in frequency and/or phase. Likewise, the second response signal can be generated by the DUT based on the second test signal, wherein it has the same frequency as the second test signal but differs from the second test signal in frequency and/or phase.

(12) The at least two test signals can be signals that travel from the RF signal generator **12** towards the port **14**, and the at least two response signals can be signals that travel from the port **14** towards the RF signal generator **12**.

(13) The couplers **16**, **17** can be directional couplers, i.e., couplers which only couple power flowing in one direction. The forward coupler **16** can be configured to couple defined amounts of the electromagnetic power of signals, which travel on the signal path **13** from the RF signal generator **12** towards the port **14**, to the measurement unit **15**. Likewise, the reverse coupler **17** can be configured to couple defined amounts of the electromagnetic power of signals, which travel on the signal path **13** from the port **14** towards the RF signal generator **12**, to the measurement unit **15**. As such, the couplers **16**, **17** can be configured to forward at least a part of the electromagnetic power of the test signals respectively the response signals to the measurement unit **15**.

(14) By supplying the DUT with two (or more) test signals in parallel, the measurement device **10** can measure a response of the DUT to the two frequency signals simultaneously for one port. Due to the frequency offset of the test signals, the overall measurement time can be reduced, e.g. cut in half if two test signals are used in parallel. In particular for passive DUTs **11**, such as cables or electrical lines, the measurement time can be reduced significantly (e.g., by a factor of two).

(15) The RF signal generator **12** can be configured to stepwise or continuously amend the frequencies of the at least two test signals by certain frequency steps over a determined frequency bandwidth. The frequency steps can be predefined and/or adjustable. In this way, the RF signal generator **12** can perform a frequency sweep over the frequency bandwidth. During each step of the frequency sweep, the RF signal generator **12** generates at least two RF test signals (at different frequencies). By generating (and measuring) multiple test signals in parallel, for each step of the frequency sweep, fewer measuring steps are required and the measuring time required to cover a determined frequency bandwidth at a certain frequency resolution can be reduced.

(16) Preferably, during the entire frequency sweep, the frequency of the first test signal is always lower than the frequency of the second test signal which is generated simultaneously to the first test signal. For example, at the start of the frequency sweep the frequency of the first test signal (f1) is 20 MHz and the frequency of the second test signal (f2) is 25 MHz, and at the end of the frequency sweep f1=995 MHz and f2=1,000 MHz. An increase of f1 and f2 during the sweep can be done synchronously by a few Hz or a few kHz each step. This is especially useful when the signals are subsequently fed to an ADC, because both frequencies are close together and can be processed together by the same ADC.

(17) For instance, the distance between both frequencies should be smaller than a bandwidth of the ADC. In an example, the distance between both frequencies f1 and f2 is smaller than 50%, 40%, 30%, 25%, 20%, 15% or 10% of the bandwidth of the ADC. Furthermore, the distance between both frequencies can be bigger than 1%, 3%, 5% or 10% of the bandwidth of the ADC.

(18) For example, the frequencies of the first and the second test signal can always be changed at the same time during the frequency sweep. Both signals and, thus, both frequencies could be generated by different RF sources which are synchronous to each other.

(19) Furthermore, during the frequency sweep, the frequencies of the first test signal can be predominately lower than the frequency of the second test signal. For instance, f1 is swept from 0 to 500 MHz while f2 is swept from 500 to 1000 MHz.

(20) For instance, during the frequency sweep, the two test signals are increased by the same

frequency starting from their respective initial frequencies.
(21) Furthermore, at an initial time t=0 of the frequency sweep, the frequency of the first test signal can be lower than the frequency of the second test signal, and at time t=1 (the next frequency step) the frequency of the first test signal can be higher than the frequency of the second test signal att=0 (e.g., f1(t=0)<f2 (t=0); f1(t=1)>f2 (t=0); and f1(t=1)<f2 (t=1)). Thus, the frequencies of the first and the second test signal can be increased in a staggered manner, for instance:
(22) TABLE-US-00001 t = 0: f1 = 100 MHz; f2 = 101 MHz t = 1: f2 = 102 MHz; f2 = 103 MHz.
(23) In another example, the frequencies of both test signals are swept in a blockwise manner. For instance, f1 is swept from 0 MHz to 25 MHz and f2 is swept from 25 MHz to 50 MHz, or f1 is swept from 50 MHz to 75 MHz and f2 is swept from 75 MHz to 100 MHz. One frequency block can comprise a certain frequency bandwidth in which only one test signal is present at a time. In other words, each frequency block of a test signal is free of the other test signal.
(24) The exemplary device **10** shown in FIG. **1** comprises a further port **14′** and a further signal path **13′** which connects the further port **14′** to the RF signal generator **12**. In the following, the signal path **13** and the port **14** are referred to as first signal path **13** and first port **14**, and the further signal path **13′** and further port **14′** are referred to as second signal path **13′** and second port **14′**, respectively. However, this does not imply that the device **10** may only comprises two signal paths or two ports. In fact, the device **10** may comprise at least two further signal path connected to at least two further ports, each signal path being coupled to the measurement unit and the RF signal generator in a similar fashion as the signal paths **13** and **13′**.
(25) As shown in FIG. **1**, the DUT **11** can be connected to the first port **14** and to the second port **14′**.
(26) The second signal path **13′** can be configured to receive the at least two test signals from the RF signal generator **12** and to feed said test signals to the second port **14′** for forwarding to the DUT **11** and/or to receive at least two further response signals of the DUT from the second port **14′**.
(27) In particular, the second signal path **13′** (and the second port **14′**) can receive the at least two further response signals in response to the test signals being forwarded to the DUT via the first port **14** or via the second port **14′**. In case the test signals are forwarded via the first port **14**, the response signal received at the second port **14′** can be “transmitted” signals (i.e., generated by the transmission of the test signals through the DUT **11**), and in case the test signals are forwarded to the DUT **11** via second port **14′**, the response signal received at the second port **14′** can be “reflected” signals (i.e., generated by the reflection of the test signals from the DUT **11**).
(28) Likewise, the first signal path **13** (and the first port **14**) can receive the at least two response signals in response to the test signals being forwarded to the DUT via the first port **14** or via the second port **14′**.
(29) The device can comprise a further (or second) forward coupler **16′** being connected to the second signal path **13′** and adapted to forward at least a part of each of the at least two test signals to the measurement unit **15** and/or a further (or second) reverse coupler **17′** being connected to the further signal path and adapted to forward at least a part of each the at least two further response signals to the measurement unit **15**.
(30) In other words, the measurement device **10** can simultaneously forward at least two test signals via the first port **14** or the second port **14′** to the DUT and simultaneously receive at least two response signals to the test signals at the first port **14** and/or at the second port **14′**. In each case, the measurement device **10** can forward respective parts of the at least two test signals and the at least two response signals to the measurement unit **15** which can simultaneously measure these forwarded signal parts in amplitude and phase.
(31) The measurement device **10** can be operated in a first operating state or mode and in a second operating state or mode.
(32) In the first operating state (or mode), the measurement device **10** can be configured to generate and transmit the at least two test signals through the first signal path **13** towards the first port **14**;

wherein the measurement device **10** is configured to: a) forward the parts of each of the at least two test signals via the first forward coupler **16** to the measurement unit **15**; b) forward the parts of each of the at least two response signals from the first port **14** via the first reverse coupler **17** to the measurement unit **15**; and c) forward the parts of each of the at least two further response signals from the second port **14'** via the second reverse coupler **17'** to the measurement unit **15**.

(33) In the second operating state (or mode), the measurement device **10** can be configured to generate and transmit the at least two test signals through the second signal path **13'** towards the second port **14'**; wherein the measurement device **10** is configured to: a) forward the parts of each of the at least two test signals via the second forward coupler **16'** to the measurement unit **15**; b) forward the parts of each of the at least two response signals from the first port **14** via the first reverse coupler **17** to the measurement unit **15**; and c) forward the parts of each of the at least two further response signals from the second port **14'** via the second reverse coupler **17'** to the measurement unit **15**.

(34) In both the first and second operating state, the measurement unit **15** can be adapted to simultaneously measure the forwarded parts of the at least two test signals, the at least two response signals and/or the at least two further response signals, each in amplitude and phase.

(35) The measurement unit **15** can be configured to calculate S-parameters of the DUT **11** over a predefined or determined frequency bandwidth. This calculation can be at least partially based on the amplitude and phase measurements of the parts of the at least two test signals, the at least to response signals and/or the at least two further response signals.

(36) FIG. **2** shows a schematic diagram of the measurement device **10** according to an embodiment. In particular, FIG. **2** shows elements of the measurement device **10** which are coupled between a signal path **13** and the measurement unit **15**.

(37) The signal path **13** in FIG. **2** is connected to two RF sources **12***a*, **12***b* of the RF signal generator **12**. For instance, each of the RF source **12***a*, **12***b* can be configured to generate one of two test signals. A direction coupler **21** is connected to the signal path. The directional coupler **21** can comprise the forward coupler **16** and the reverse coupler **17**.

(38) The forward coupler **16** and the reverse coupler **17** can each be connected to a mixing unit which is configured to down-convert the forwarded parts of the two test signals respectively the forwarded parts of the two response signals into IF signals.

(39) In the example shown in FIG. **2**, each of the mixing units comprises a single mixer **23**, wherein the mixers **23** are connected to a common source **22** (e.g., a local oscillator). The mixers **23** can be frequency mixers for RF signals. For instance, the mixer **23** coupled to the forward coupler **16** receives and down-converts the decoupled parts of the two test signals and the mixer **23** coupled to the reverse coupler **17** receives and down-converts the decoupled parts of the two response signals.

(40) Each of the mixers **23** can forward the generated IF signals to a respective ADC **24**. The ADCs **24** can be configured to digitalize the IF signals forwarded by the mixers **23**. Each of the ADCs **24** can be a broadband ADC capable of digitalizing both down-converted test signals respectively both down-converted response signals in parallel.

(41) Each of the ADCs **24** can be further connected to two digital mixers **25**. For instance, each of the digital mixers **25** is configured to receive one of the digitalized test signals respectively one of the digitalized response signals, e.g. to further down convert said signals. In each case, two of the mixers **25** can each be coupled to a common numerically controlled oscillator **26**. Each digital mixer **25** can feed its output signal to a respective digital filter **27**. After being filtered by the filters **27**, the signals can be forwarded to the measurement unit **15** (not shown in FIG. **2**).

(42) FIG. **3** shows a further example for the elements of the measurement device **10** which are coupled between the signal path **13** and the measurement unit **15**.

(43) In the example shown in FIG. **3**, the mixing unit coupled to the forward coupler **16** and the mixing unit coupled to the reverser coupler **17** each comprise two mixers **23**. For instance, each

mixer **23** receives and down-converts a part of one of the two test signals or of one of the two response signals, respectively.
(44) The mixers **23** can be coupled to two respective sources **22** (e.g., local oscillators). For instance, a first source **22** is coupled to the mixers **23** which down-convert the parts of the first test signal and the first response signal (that is based on the first test signal), and a second source **22** is coupled to the mixers **23** which down-converts the parts of the second test signal and the second response signal (that is based on second test signal).
(45) Each of the mixers **23** in FIG. **3** can be coupled to a respective ADC **24** for digitalizing the down-converted IF signals. Each ADC **24** can further forward the digitalized signal to a digital mixer **25** coupled to a numerically controlled oscillator **26**. Each digital mixer **25** can feed its output signal to a respective digital filter **27**. After being filtered by the filters **27**, the signals can be forwarded to the measurement unit **15** (not shown in FIG. **2**).
(46) In case a frequency sweep is performed, the local oscillator(s) **22** of the respective mixers **23** can be adapted according to the actual frequency of the at least two test signals.
(47) For instance, the minimal frequency offset between the two (or more) test signals can depend on the filter **27**. In particular, the frequency offset can be chosen such that the signals can be separated by the filter stage which can be formed by the filters **27** in FIG. **2** or **3**.
(48) For instance, each signal path **13**, **13′** of the measurement device **10** that connects the RF signal generator **12** to a port can be coupled to the measurement unit **15** as shown in FIG. **2** or FIG. **3**.
(49) FIGS. **4**A-**4**B show schematic diagrams of the measurement device **10** according to an embodiment.
(50) Thereby, FIG. **4**A shows that the measurement device **10** can comprise four ports P**1**-P**4**, wherein the DUT **11** can be connected to each of the four ports. FIG. **4**B shows that each of the ports P**1**-P**4** can be connected to the RF signal generator **12** via a respective signal path, wherein a directional coupler **21** (which comprises the forward coupler **16** and the reverse coupler **17**) can forward parts of at least two test signals and at least two response signals traveling on the signal path to the measurement unit **15**.
(51) With the setup shown in FIGS. **4**A and **4**B, 4-port characterization measurements of the DUT **11** can be carried out with multiple (at least two) test signals in parallel. In this way, the measurement times of the 4-port measurements can be strongly reduced (at least by a factor of two).
(52) FIG. **5** shows a schematic diagram of a system **50** comprising a measurement device **10** according to an embodiment. The system **50** further comprises a switch matrix **51** which is connected between at least one port **14** of the measurement device and the DUT **11**. Thereby, the measurement device **10** can be configured to control this switch matrix **51**. For instance, the measurement device **10** can be connected to the switch matrix **51** via a suitable communication interface, e.g. USB or LAN. The switch matrix **51** can comprise semiconductor switches or mechanical switches, the switching operation of which can controlled by the measurement device **10**.
(53) For instance, the switch matrix **51** can be connected to four ports **14** of the measurement device **10** and can provide a large number of connecting ports, e.g. 32 ports, for connecting the DUT **11**. The DUT **11** can be a cable, such as an Ethernet cable which comprises 8 twisted conductor pairs, i.e., in total 16 wires which are to be measured. For characterization, both ends of the cable can be connected to the switch matrix **51** and, thus, the measurement device **10**. The switch matrix **51** can then successively connect both ends of the twisted conductor pairs to the ports of the measurement device **10** to perform 4-port measurements.
(54) By using such a switching matrix **51**, individual screw-on steps can be eliminated, so that the measurements for the different lines in the cable can be performed directly one after the other which leads to an additional reduction in measurement time. Furthermore, the parallel

measurements at two (or more) frequencies further reduces the measurement time. In this way, the throughput of cable tests can be significantly enhanced.
(55) For example, the system **50** can further comprise a connection board **52** which is switched between the switch matrix **51** and the DUT **11**. The connection board **52** can provide the correct connector or adapter for connecting the DUT **11**. The connection board **52** can form a cable fixture or a measurement adapter.
(56) FIG. **6** shows a flow diagram of a method **60** of operating a system **50** comprising a measurement device **10** and a switch matrix **51** according to an embodiment. The method **60** is thereby used to analyze a DUT, such as a cable, which comprises a plurality of conductor lines. The method **60** can be carried out with the system **50** as shown in FIG. **5**.
(57) The method **60** comprises the steps of: connecting **61** both ends of the plurality of conductor lines to respective test ports of the switch matrix **51**; forwarding **62** the at least two test signals to one end of one of the plurality of conductor lines; and measuring **63** the at least two response signals at the other end of the conductor line which receives the test signal, and at both ends of one, more or all further conductor lines of the plurality of conductor lines which are connected to the switch matrix **51**.
(58) Thereby, the switch matrix **51** can forward the test signals from the measurement device **10** to the respective ends of the conductor lines, and can forward the response signals from the respective ends of the conductor lines to the measurement device **10** for the measurements.
(59) The DUT can be a cable, in particular a cable comprising twisted wires or conductor lines (twisted pairs). For example, each two of the conductor lines of the cable are twisted together.
(60) By measuring both the transmission at the end of the conductor line which receives the test signals and at one or more other conductor lines of the cable, the attenuation of the first line and the crosstalk to the other conductor line(s) can be measured.
(61) For example, the test signals are forwarded to one end of a first conductor line and the response signals are measured at the other end of the first conductor line and at both ends of a second conductor line, wherein the first and the second conductor line are twisted together. In this way, a 4-port measurement can be carried out with a twisted pair cable.
(62) In a further step, the crosstalk to further (third, fourth etc.) conductor lines can be measured, by measuring the at least two response signals at both ends of the further conductor lines. For instance, when forwarding the test signals to one conductor line (e.g., line 1) the response signals can be measured sequentially or simultaneously at the other conductor lines, for instance: line 1.fwdarw.line 2; line 1.fwdarw.line 3; line 1.fwdarw.line 4; line 2.fwdarw.line 3; . . . .
(63) After forwarding the test signals to one end of a conductor line and measuring the response at the other end (and at further lines), the test signals can subsequently be forwarded to the other end of the conductor line and the measurement can be repeated.
(64) The measurements with multiple lines can be carried out in parallel. For instance, the test signals can be forwarded to one end of line 1 and the response can be measured at both ends of line 2. In parallel thereto, the test signals can be forwarded to one end of line 5 and the response can be measured at both ends of line 6. Per line, at least two test signals which have a different frequency can be used.
(65) For instance, the test signals forwarded to one line (e.g., line 1) can be different to the test signals forwarded to another line (e.g., line 5). In this way, unwanted crosstalk can be avoided, as a signal from line 5 can couple to line 2 but will not be measured because it is outside of the downcoverted intermediate frequency IF.
(66) All features described above or features shown in the figures can be combined with each other in any advantageous manner within the scope of the disclosure.

Claims

1. A measurement device for characterizing a device-under-test, DUT, comprising: an RF signal generator configured to generate at least two test signals in parallel, wherein the at least two test signals each have a different frequency; a signal path connecting the RF signal generator to a port of the measurement device, wherein the port is arranged for being connected to the DUT; the signal path being configured to feed the at least two test signals to the port and to receive at least two response signals of the DUT from the port; a measurement unit; a forward coupler being connected to the signal path and adapted to forward at least a part of each of the at least two test signals to the measurement unit; and a reverse coupler being connected to the signal path and adapted to forward at least a part of each of the at least two response signals to the measurement unit; the measurement unit being adapted to simultaneously measure the forwarded parts of the at least two test signals and the at least two response signals, each in amplitude and phase.
2. The measurement device of claim 1, wherein the RF signal generator is configured to amend the frequencies of the at least two test signals by predefined and/or adjustable frequency steps over a determined frequency bandwidth.
3. The measurement device of claim 2, wherein the at least two test signals are each increased by the same frequency, starting from their respective initial frequencies.
4. The measurement device of claim 1, further comprising: a first mixing unit which is configured to down-convert the forwarded parts of the at least two test signals into intermediate frequency, IF, signals; a first analog-to-digital converter, ADC, unit which is configured to digitize the down-converted parts of the test signals; a second mixing unit which is configured to down-convert the forwarded parts of the at least two response signals into IF signals; and a second ADC unit which is configured to digitize the down-converted parts of the response signals.
5. The measurement device of claim 4, wherein the first mixing unit comprises only one mixer which is configured to down-convert the forwarded parts of the at least two test signals; and wherein the first ADC unit comprises only one ADC which is configured to digitize the down-converted parts of the test signals.
6. The measurement device of claim 4, wherein the second mixing unit comprises only one mixer which is configured to down-convert the forwarded parts of the at least two response signals; wherein the second ADC unit comprises only one ADC which is configured to digitize the down-converted parts of the response signals.
7. The measurement device of claim 4, wherein the first mixing unit comprises at least two mixers, wherein a first mixer of the at least two mixers is configured to down-convert the forwarded part of a first one of the at least two test signals and wherein a second mixer of the at least two mixers is configured to down-convert the forwarded part of a second one of the at least two test signals; and wherein the first ADC unit comprises at least two ADCs, wherein a first ADC of the at least two ADCs is configured to digitize the down-converted part of the first one of the test signals and wherein a second ADC of the at least two ADCs is configured to digitize the down-converted part of the second one of the test signals.
8. The measurement device of claim 4, wherein the second mixing unit comprises at least two mixers, wherein a first mixer of the at least two mixers is configured to down-convert the forwarded part of a first one of the at least two response signals and wherein a second mixer of the at least two mixers is configured to down-convert the forwarded part of a second one of the at least two response signals; and wherein the second ADC unit comprises at least two ADCs, wherein a first ADC of the at least two ADCs is configured to digitize the down-converted part of the first one of the response signals and wherein a second ADC of the at least two ADCs is configured to digitize the down-converted part of the second one of the response signals.
9. The measurement device of claim 1, further comprising: a further signal path connecting the RF signal generator to a further port of the measurement device, wherein the further port is arranged for being connected to the DUT; wherein the further signal path is configured to feed the at least

two test signals to the further port and/or to receive at least two further response signals of the DUT from the further port.

10. The measurement device of claim 9, further comprising: a further forward coupler being connected to the further signal path and adapted to forward at least a part of each of the at least two test signals to the measurement unit; and/or a further reverse coupler being connected to the further signal path and adapted to forward at least a part of each the at least two further response signals to the measurement unit.

11. The measurement device of claim 10, wherein, in a first operating state, the measurement device is configured to transmit the at least two test signals through the first signal path to the first port; wherein the measurement device is configured to: a) forward parts of each of the at least two test signals via the forward coupler to the measurement unit; b) forward parts of each of the at least two response signals from the first port via the reverse coupler to the measurement unit; and c) forward parts of each of the at least two further response signals from the further port via the further reverse coupler to the measurement unit.

12. The measurement device of claim 10, wherein, in a second operating state, the measurement device is configured to transmit the at least two test signals through the further signal path to the further port; wherein the measurement device is configured to: a) forward parts of each of the at least two test signals via the further forward coupler to the measurement unit; b) forward parts of each of the at least two response signals from the port via the reverse coupler to the measurement unit; and c) forward parts of each of the at least two further response signals from the further port via the further reverse coupler to the measurement unit.

13. The measurement device of claim 1, wherein the measurement unit is configured to calculate S-parameters of the DUT over a determined frequency bandwidth at least based on the measurements of the forwarded parts of the at least two test signals and the at least two response signals.

14. The measurement device of claim 1, wherein the measurement device comprises at least four ports, wherein each of the at least four ports is arranged for being connected to the DUT, and wherein each of the at least four ports is connected to the RF signal generator by a respective signal path; wherein the measurement device is adapted to apply the at least two test signals to each of the at least four ports in an alternating manner; and wherein each signal path is connected to the measurement unit by a respective forward coupler and a respective reverse coupler.

15. A method of operating a system to analyze a device-under-test, DUT, the DUT comprises a plurality of conductor lines, the system comprises a measurement device and a switch matrix which is connected between at least one port of the measurement device and the DUT, wherein the measurement device is configured to control the switch matrix, and the measurement device comprises: an RF signal generator configured to generate at least two test signals in parallel, wherein the at least two test signals each have a different frequency; a signal path connecting the RF signal generator to a port of the measurement device, wherein the port is arranged for being connected to the DUT; the signal path being configured to feed the at least two test signals to the port and to receive at least two response signals of the DUT from the port; a measurement unit; a forward coupler being connected to the signal path and adapted to forward at least a part of each of the at least two test signals to the measurement unit; and a reverse coupler being connected to the signal path and adapted to forward at least a part of each of the at least two response signals to the measurement unit; the measurement unit being adapted to simultaneously measure the forwarded parts of the at least two test signals and the at least two response signals, each in amplitude and phase, the method comprising the steps of: connecting both ends of the plurality of conductor lines to respective test ports of the switch matrix; forwarding the at least two test signals to one end of one of the plurality of conductor lines; and measuring the at least two response signals: at the other end of the conductor line which receives the at least two test signals, and at both ends of one, more or all further conductor lines of the plurality of conductor lines.

16. The method of claim 15, wherein the switch matrix comprise a plurality of test ports adapted to

be connected to one or more than one DUT; wherein the switch matrix is configured to subsequently connect each of the plurality of the test ports to the at least one port of the measurement device.